

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

07

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**C****C. 640 cm^3**

Choice C is correct. The total volume of the cylindrical can is found by multiplying the area of the base of the can, 75 cm^2 , by the height of the can, 10 cm, which yields 750 cm^3 . If the syrup needed to fill the can has a volume of 110 cm^3 , then the remaining volume for the pieces of fruit is $750 - 110 = 640 \text{ cm}^3$.

Choice A is incorrect because if the fruit had a volume of 7.5 cm^3 , there would be $750 - 7.5 = 742.5 \text{ cm}^3$ of syrup needed to fill the can to the top. Choice B is incorrect because if the fruit had a volume of 185 cm^3 , there would be $750 - 185 = 565 \text{ cm}^3$ of syrup needed to fill the can to the top. Choice D is incorrect because it is the total volume of the can, not just of the pieces of fruit.

Q2**63**

The correct answer is 63. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is 27° . It follows that the measure of angle D is also 27° . It's given that angle E is a right angle. Therefore, the measure of angle E is 90° . Let x represent the measure, in degrees, of angle F . Since the measures of the angles in a triangle sum to 180° , it follows that $27 + 90 + x = 180$, or $117 + x = 180$. Subtracting 117 from both sides of this equation yields $x = 63$. Therefore, the measure, in degrees, of angle F is 63.

Q3**31**

The correct answer is 31. The circumference of a circle is equal to $2\pi r$ centimeters, where r represents the radius, in centimeters, of the circle, and the diameter of the circle is equal to $2r$ centimeters. It's given that a circle has a circumference of 31π centimeters. Therefore, $31\pi = 2\pi r$. Dividing both sides of this equation by π yields $31 = 2r$. Since the diameter of the circle is equal to $2r$ centimeters, it follows that the diameter, in centimeters, of the circle is 31.

Q4**57**

The correct answer is 57. Based on the figure, the angle with measure y° and the angle vertical to the angle with measure 58° are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure 58° also has measure 58° . It's given that lines q and r are parallel. Therefore, same side interior angles between lines q and r are supplementary. It follows that $y + 58 = 180$. If $y = 2x + 8$, then the value of x can be found by substituting $2x + 8$ for y in the equation $y + 58 = 180$, which yields $2x + 8 + 58 = 180$, or $2x + 66 = 180$. Subtracting 66 from both sides of this equation yields $2x = 114$. Dividing both sides of this equation by 2 yields $x = 57$. Thus, if $y = 2x + 8$, the value of x is 57.

Q5**4205**

The correct answer is 4,205. The exterior surface area of a figure is the sum of the areas of all its faces. It's given that the box does not have a lid and that each side of the box is in the shape of a square. Therefore, the box consists of 5 congruent square faces. It's also given that the length of each edge is 29 inches. Let s represent the length of an edge of a square. It follows that the area of a square is equal to s^2 . Therefore, the area of each of the 5 square faces is equal to 29^2 , or 841, square inches. Since the box consists of 5 congruent square faces, it follows that the sum of the areas of all its faces, or the exterior surface area of this box without a lid, is 5841, or 4,205, square inches.

Q6**B****B. 17.5**

Choice B is correct. Since $\overline{AF}$, $\overline{BE}$, and $\overline{CD}$ are parallel, quadrilaterals $AFEB$ and $BEDC$ are similar. Let x represent the length of $\overline{ED}$. With similar figures, the ratios of the lengths of corresponding sides are equal. It follows that $\frac{9}{18.5} = \frac{8.5}{x}$. Multiplying both sides of this equation by 18.5 and by x yields $9x = (18.5)(8.5)$, or $9x = 157.25$. Dividing both sides of this equation by 9 yields $x = 17.47$, which to the nearest tenth is 17.5.

Choices A, C, and D are incorrect and may result from errors made when setting up the proportion.

Q7**4.41**

Accept also: 441/100

The correct answer is 4.41. The area, A , of a circle is given by the formula $A = \pi r^2$, where r is the radius of the circle. It's given that the area of the circle is $b\pi$ square inches, where b is a constant, and the radius of the circle is 2.1 inches. Substituting $b\pi$ for A and 2.1 for r in the formula $A = \pi r^2$ yields $b\pi = \pi 2.1^2$. Dividing both sides of this equation by π yields $b = 4.41$. Therefore, the value of b is 4.41.

Q8**D****D. 32**

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z , respectively. Therefore, side AB corresponds to side XY . Since the length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC , it follows that the measure of side XY is 2 times the measure of side AB . Thus, since the measure of side AB is 16, then the measure of side XY is 216, or 32.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure of side AB , not side XY .

Choice C is incorrect and may result from conceptual or calculation errors.

Q9**29**

The correct answer is 29. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. Since the base of the cylinder is a circle with radius r , the area of the base of the cylinder is πr^2 . It's given that a right circular cylinder has a volume of 377 cubic centimeters; therefore, $V = 377$. It's also given that the area of the base of the cylinder is 13 square centimeters; therefore, $\pi r^2 = 13$. Substituting 377 for V and 13 for πr^2 in the formula $V = \pi r^2 h$ yields $377 = 13h$. Dividing both sides of this equation by 13 yields $29 = h$. Therefore, the height of the cylinder, in centimeters, is 29.

Q10

A

A. $\frac{350}{15}$

Choice A is correct. The figure shows that angle P in $\triangle QPR$ and angle T in $\triangle STR$ are right angles. It follows that angle P is congruent to angle T . The figure also shows that the measures of angle QRP and angle SRT are both x° . Therefore, angle QRP is congruent to angle SRT . It's given that $\triangle QPR$ is similar to $\triangle STR$. Since angle P is congruent to angle T , and angle QRP is congruent to angle SRT , it follows that $\bar{QR}$ corresponds to $\bar{SR}$, and $\bar{QP}$ corresponds to $\bar{ST}$. Since corresponding sides of similar triangles are proportional, it follows that $\frac{SR}{QR} = \frac{ST}{QP}$. It's also given that the lengths of $\bar{ST}$, $\bar{QP}$, and $\bar{QR}$ are 14, 15, and 25, respectively. Substituting 14 for ST , 15 for QP , and 25 for QR in the equation $\frac{SR}{QR} = \frac{ST}{QP}$ yields $\frac{SR}{25} = \frac{14}{15}$. Multiplying each side of this equation by 25 yields $SR = \frac{14}{15}25$, or $SR = \frac{350}{15}$. Thus, the length of $\bar{SR}$ is $\frac{350}{15}$.

Choice B is incorrect. This is the result of solving the equation $\frac{SR}{25} = \frac{14}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice C is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice D is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{25}$, not $\frac{SR}{25} = \frac{14}{15}$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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